

Filter Monitors

1. If you wish to include individual monitors by ID, enter the IDs here, separated by commas. If you do not enter any monitors here, all monitors which meet the rest of the selection criteria will be included:

2. If you wish to exclude individual monitors by ID, enter the IDs here, separated by commas:

3. If you wish to restrict monitors to certain states or areas, enter comma separated state abbreviations, and minimum and/or maximum latitudes and longitudes:

States:

Minimum Longitude:

Maximum Longitude:

Minimum Latitude:

Maximum Latitude:

4. Select maximum POC code to include and the POC preference:

Maximum POC:

POC Preference Order:

5. Select the Methods you wish to include:

☒ 116
☒ 117
☒ 118
☒ 119
☒ 120
☐ 785

6. Select the parameters specific to the pollutant:

Number of Valid Observations Required Per Quarter:

Data Types To Use
☒ Local ☐ Standard ☐ Both

Preferred Type
☒ Local ☐ Standard

Output Type
☒ Local ☐ Standard

Note that the first five options are essentially the same for each pollutant, and the sixth option depends on the pollutant. The example above shows what the form looks like with **PM2.5** as the selected **Pollutant**.

1. Include specific monitors. Here you can specify particular monitor IDs that you want to include in your analysis. If this is left blank, then BenMAP-CE will include all monitors that meet the rest of the selection criteria.

2. Exclude specific monitors. Here you can exclude any particular monitor IDs from your analysis. Here again, if this option is left blank then BenMAP-CE will include all monitors that meet the rest of the selection criteria.

3. Restrict to particular states and/or latitude/longitude. You can choose monitors to include from particular states, by listing the two-character state abbreviation (e.g., CA = California). You can also choose monitors within particular latitude and longitude ranges. The default values for latitude (20 to 50) and longitude (-130 to -65) completely

include the continental U.S. Here again, if this option is left blank then BenMAP-CE will include all monitors that meet the rest of the selection criteria.

4. Parameter Occurrence Code (POC)²¹. Sometimes, multiple monitors are collocated at the same site measuring the same parameter (e.g., to check precision). The **Maximum POC** specifies the highest POC value allowed in the data. The default is a value of 4. And to choose one monitor when more than one monitor is in the same location, the POC Preference Order specifies the preferred ordering of POC codes.

5. Methods.²² The Method codes listed indicate U.S. EPA-defined methods for collecting and analyzing samples; these codes depend on the pollutant. In the case of PM_{2.5}, only federal-reference methods (FRM) are chosen by default -- specifically numbers 116 through 120. In the case of Ozone (O₃), all methods are checked by default.

6. Parameters Specific to the Pollutant. The default options vary by pollutant. Below, we have described the options that appear with *PM_{2.5}* and *ozone*.

- **PM_{2.5} Monitor Filter:** The **Number of Valid Observations Required per Quarter** specifies the number of days of data needed. The default is to require 11 observations per quarter. The **Data Types to Use** options specify whether to use data at *Local* conditions (parameter code 88101), *Standard* conditions (parameter code 81104), or *Both*.²³ The default for PM_{2.5} is to use data at *Local* conditions. When data at standard and local conditions are both available at the same monitor location, the **Preferred Type** allows you to choose which to use – the default is *Local*. The **Output Type** option is designed to allow you to make the data reasonably consistent when both local and standard condition data are used. The default is to use the *Local* output type, so *Standard* condition data will be converted to *Local*.
- **Ozone Monitor Filter:** The *ozone* specific options differ from *PM_{2.5}* in large part because ozone is monitored hourly in the United States. The **Number of Valid Hours** specifies the number of hours needed for a particular day of monitoring to be considered “valid.” BenMAP-CE counts the number of non-missing hourly values from the **Start Hour** through the **End Hour** and compares this number with that specified in the **Number of Valid Hours**.

The **Percent of Valid Days** specifies the percent of days between the **Start Date** and the **End Date** that need to be valid for the monitor itself to be considered valid. The

²¹ For information about data codes used in U.S. EPA’s Air Quality System, refer to the *AQS Data Coding Manual*: https://www.epa.gov/sites/production/files/2015-09/documents/aqs_data_coding_manual_0.pdf

²² For more information, refer to U.S. EPA’s *List of Designated Reference and Equivalent Methods*: <https://www3.epa.gov/ttnamti1/files/ambient/criteria/AMTIC%20List%20Dec%202016-2.pdf>

²³ Particulate concentrations are expressed in either local conditions (volume is at temperature and pressure of the ambient sample) or at standard conditions (where the volume has been converted to standard conditions, typically 20° C at 760 mm Hg).

default is 50 percent of the days between May 1 and September 30. The example below shows what the form looks like with *Ozone* as the selected **Pollutant**.

You can view a map of your data with the specified filter options by clicking the **Map** button on the bottom left side of the **Filter Monitors** window. This provides a brief preview of what will be shown once the data are loaded.

You can also export your data by clicking the **Export** button, which is found in the bottom left side of the **Filter Monitors** window. A **Save As** window will appear, allowing you to save the data as a CSV file.

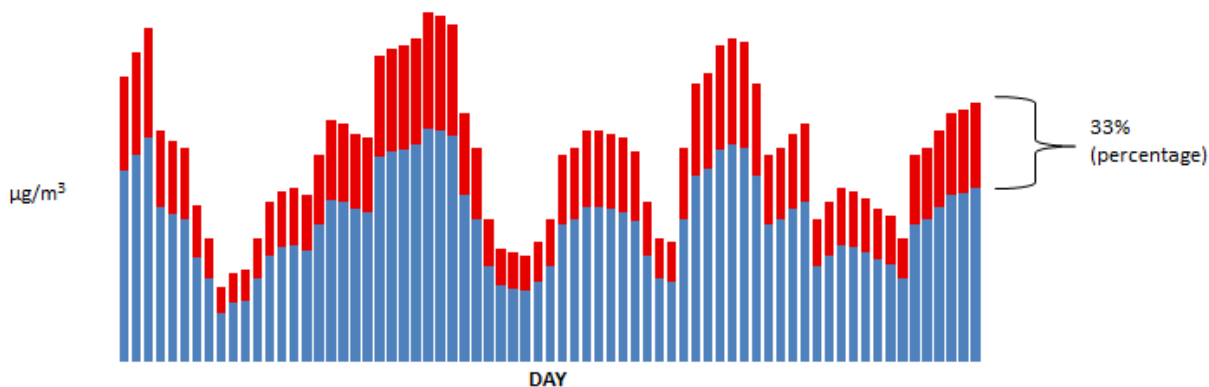
5.3 Monitor Rollback

The *Monitor Rollback* option allows you to quickly test what the benefits would be from reducing historical monitor levels. BenMAP-CE has three methods to reduce, or "roll back," monitor data: *Percentage Rollback*, *Incremental Rollback*, or *Rollback to a Standard*. Each of these methods is depicted below. Note that with each of these methods you can use the same two interpolation algorithms (*Closest Monitor* or *VNA*) as you can use with *Monitor Data*.

Fundamental Concept - Rollback

A **rollback** is a simplified type of BenMAP analysis in which all Baseline scenario monitor data are reduced in the Control scenario using a single, uniform rule. BenMAP-CE offers three rollback options: **Percentage rollback** reduces all monitor observations by the same percentage. **Incremental rollback** reduces all observations by the same increment. **Rollback to a standard** reduces monitor concentrations as necessary to ensure all monitor observations are at or below a user-specified air quality standard.

Percentage Rollback reduces all monitor observations by the same percentage:



Incremental Rollback reduces all observations by the same increment:

